

First Terminal Examination - 2080

Class : XI

Subject: Chemistry (3011)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- Any 2p orbital can accommodate maximum number of electrons are
a. 2 b. 6 c. 10 d. 14
- The valance electron of an atom of a d-block element has $n = 4$, $l = 0$, $m = 0$, $s = +\frac{1}{2}$, the atomic number is:
a. 11 b. 24 c. 19 d. 26
- There are three energy level of an atom. How many spectral lines are possible in its emission spectra?
a. 1 b. 2 c. 3 d. 4
- Which of the following set of quantum number is not possible for 4p-electron?
a. $n = 4$, $l = 1$, $m = -1$, $s = +\frac{1}{2}$ b. $n = 4$, $l = 1$, $m = 0$, $s = +\frac{1}{2}$
c. $n = 4$, $l = 1$, $m = 2$, $s = +\frac{1}{2}$ d. $n = 4$, $l = 1$, $m = -1$, $s = -\frac{1}{2}$
- The adsorption of hydrogen by Pd is
a. reduction b. occlusion

c. hydrogenation

d. dehydrogenation

6. Ozone can be detected by using

a. Na

b. Ag

c. Hg

d. Ar

7. In Haber's process for the manufacture of ammonia, the catalyst used is

a. Finely divided nickel

b. Finely divided molybdenum

c. Finely divided iron

d. Finely divided platinum

8. There are huge numbers of organic compound in existence. This is due to the

a. Tetra-covalency of carbon

b. Catenation property of carbon

c. Isomerism

d. Small size of carbon

9. The third member of an alkene is

a. Propene

b. Butene

c. Ethene

d. Pentene

10. The IUPAC name of
$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{OH} \\ | \\ \text{CH}_3 \end{array}$$

a. butan-2-ol

b. 2-methylpropanol

c. 2-methylpropan-2-ol

d. Butanol

11. Gasoline is the mixtures of hydrocarbon containing carbon number

a. C_3 - C_5

b. C_4 - C_7

c. C_6 - C_{11}

d. C_{10} - C_{15}

Group - B

Give short answer to the following questions.

[8 × 5 = 40]

1. Bohr's atomic theory is able to explain the stability of atoms as well as the formation of hydrogen spectra.
 - a. Write the important postulates of the Bohr atomic theory. [3]
 - b. Draw a well labelled diagram to show different spectra series of hydrogen atom. [2]
2. Rutherford's nuclear model of atom is based on alpha particle scattering experiment.
 - a. Which experiment led Rutherford to conclude that [2]
 - (i) atom consist positively charged mass.
 - (ii) atom is mostly hollow.
 - b. Write limitations of Rutherford's nuclear model of atom. [1]
 - c. How is the formation of hydrogen spectra explained on the basis of Bohr's theory? [2]
3. Different models were put forward to explain the wave nature of an electron.
 - a. State Heisenberg Uncertainty principle. Is this theory applied to macroscopic particles like bullets? [2]
 - b. Bohr's theory violates Heisenberg's uncertainty principle. Justify. [1]
 - c. Differences between orbit and orbital. [2]
4. Hydrogen is the simplest, lightest, and most abundant element in the universe, making up more than 90 percent of all matter.
 - a. Distinguish between atomic hydrogen and nascent hydrogen. [2]

- b. Name the isotopes of hydrogen. In what respect do they differ from one another? [2]
- c. What is heavy water? Write its uses. [1]
5. Ozone is an irritating, pale blue gas that is toxic even at low concentrations.
- a. How does the formation of Ozone take place in the stratosphere? [1]
- b. How does CFC deplete the Ozone layer? [2]
- c. Write the structure of ozone. [2]
6. Urea is much demanded fertilizer in agriculture like Nepal. One of the raw materials for the manufacture of urea is ammonia.
- a. Write the principle reaction involved in the manufacture of ammonia by Haber's process. [1]
- b. Draw a flow sheet diagram for the manufacture of ammonia by Haber's process. [2]
- c. What happens when (i) ammonia solution is passed into FeCl_3 solution and (ii) ammonia is treated with mercurous nitrate paper [2]
7. Analyze the following table of homologous series of aldehydes and answer the following questions. [1+1+2+1]

methanal	ethanal	X	butanal	Y
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- a. Define homologous series.
- b. Write any two characteristics of a homologous series.
- c. Write the structure of X and Y with their IUPAC name.
- d. Homologues are never isomers, give a reason.

8. Petroleum is often termed liquid gold. The quality of diesel can be expressed in terms of cetane number. [1+1+1+2]
- Define cetane number.
 - A fuel has cetane number 78. What does it mean?
 - Why n-heptane has been assigned octane number of zero?
 - Define pyrolysis. Starting from n-hexane, how would you prepare benzene?

Group - C

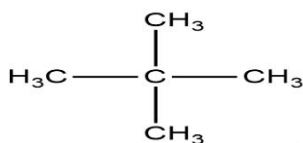
Give long answer to the following questions. [3 × 8 = 24]

9. Each atomic orbital is characterized by three quantum numbers- principal, azimuthal, and magnetic.
- Define Quantum number. Explain the quantum number which specifies the shape of an orbital. [2]
 - The first shell of an atom can't accommodate more than two electrons. Justify the statement in the light of the Pauli exclusion principle. [1]
 - State Aufbau principal. Arrange the atomic orbital according to the increasing order of energy showing (n + l) rule:
4f, 5d, 6s, 3p [2]
 - An element 'X' has 2 electrons in the K-shell, 8 in the L-shell, and 11 in the M-shell. Find out the following:
 - Electronic configuration of an atom and unpaired electron in 'X'. [1]
 - Find the electronic configuration of ion X^{+++} . Also, find all four quantum numbers of valence electrons of it. [2]

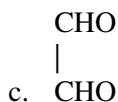
10. Oxygen is the third most abundant element by mass which readily forms oxides with other elements. Some of the oxides are given below

Na_2O	Al_2O_3	CO	SO_2	Fe_3O_4	H_2O_2
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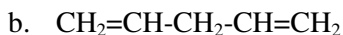
- Identify the acidic oxide, basic oxide, neutral oxide and mixed oxide from the above table with suitable reaction. [4]
 - Write two chemical equations to prove that the particular oxide is amphoteric in nature. [1]
 - How oxygen is used for medical and industrial purpose? [2]
 - Write any two applications of hydrogen peroxide. [1]
11. a. Define functional group. Write the functional group and an example of following families group. [1+2]
- ester
 - amide
- Write any two significance of functional group. [1]
 - Write the IUPAC name of the following compounds. [4]



a.



c.



❧ - The End - ❧